

 Overview

UNIT I —

▼ MATHEMATICAL
FOUNDATIONS

- ☐ 1 Calculus for Data Science
- ☐ 2 Linear Algebra
- ☐ 3 Probability

▼ UNIT II — STATISTICS

- ☐ 4 Descriptive Statistics & EDA
- ☐ 5 Sampling & Statistical Inference
- ☐ 6 Hypothesis Testing

UNIT III —

▼ PROGRAMMING & DATA
MANAGEMENT

- ☐ 7 Python Foundations
- ☐ 8 SQL & Relational Databases
- ☐ 9 Databases & Data Storage
- ☐ 10 Data Wrangling & Data Quality
- ☐ 11 Data Visualization

▼ UNIT IV — MACHINE
LEARNING CORE

- ☐ 12 The Machine Learning Workflow
- ☐ 13 Model Validation Deep Dive
- ☐ 14 Linear Regression
- ☐ 15 Logistic Regression
- ☐ 16 Evaluating Classifiers
- ☐ 17 Naive Bayes

DATA SCIENCE CURRICULUM

A Structured Academic Curriculum

Compiled from your coursework — MIS 761 (Quantitative Methods), DA 621 / ITE 621–622 (Programming for Data Analytics), ITE 451/651 (Big Data & Web Databases), and MIS 769 (Big Data Analytics for Business). Seven units, thirty-two modules, in prerequisite order. Each module covers **what it is**, **why & when**, **how it's applied** (tied to your assignments), and **the math**. Click a module to study it; mark modules complete to track progress — your progress is saved.

Unit I — Mathematical Foundations

Modules 1–3 · 0/3 complete

Unit II — Statistics

Modules 4–6 · 0/3 complete

Unit III — Programming & Data Management

Modules 7–11 · 0/5 complete

Unit IV — Machine Learning Core

Modules 12–22 · 0/11 complete

Unit V — Unsupervised Learning

Modules 23–25 · 0/3 complete

Unit VI — Big Data & Distributed Computing

Modules 26–27 · 0/2 complete

Unit VII — Text, NLP & LLMs

Modules 28–32 · 0/5 complete